SimpleVM

A SIMPLE TO USE CLOUD COMPUTING ENVIRONMENT FOR THE LIFE SCIENCES

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Granting access to the full functionality of OpenStack's Horizon Dashboard can overwhelm inexperienced users when accessing cloud infrastructures.

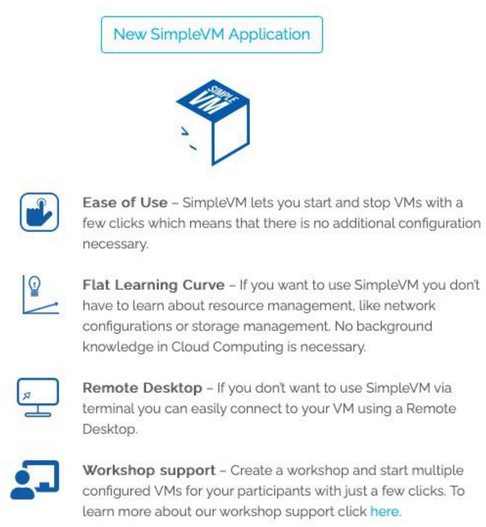
Therefore, we developed SimpleVM as one of two project types available to users within the federated de.NBI Cloud.

Based on OpenStack and interacting with its API, SimpleVM provides easy Life Science AAI-granted access to VMs for basic data-processing use-cases, but can also serve sophisticated, cluster-based computing and workshop scenarios.

Project Application

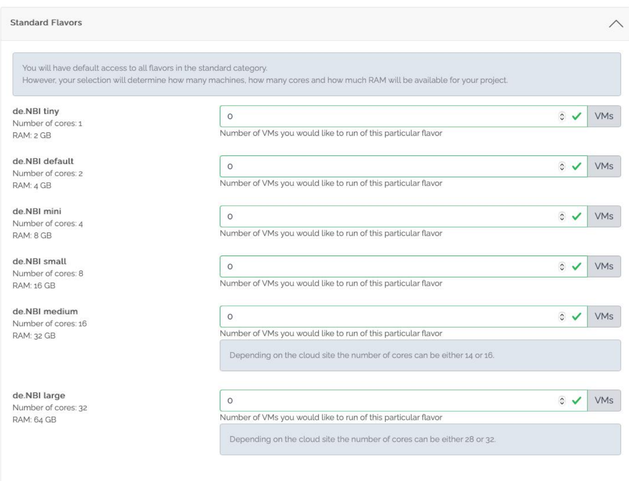
LS LOGIN

- Applicants log in via Life Science Login
- Applicants select project type, i.e., SimpleVM or OpenStack application
- Applicants enter project name a description of project goals, tools planned to be used and references to methods
- Applicants enter desired lifetime



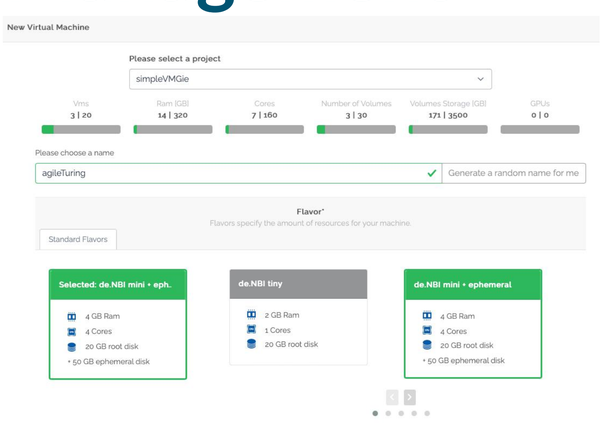
Cloud Resources

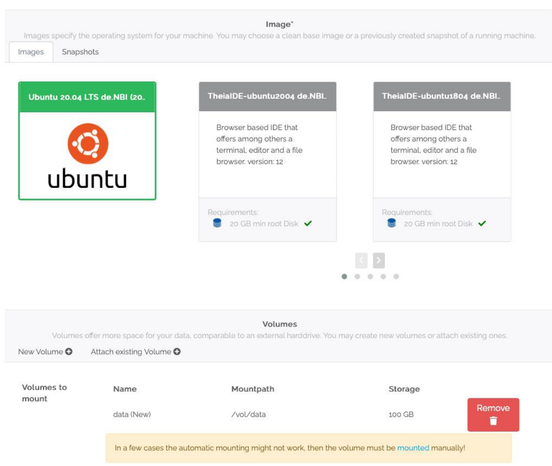
- VM flavors include
  - **Standard**
    1-32 cores
    2-64 GB RAM
  - **High Memory**
    16-32 cores
    128-512 GB RAM
  - **GPU (NVIDIA)**
    A100, V100, T4
    16-32 cores
    64-256 GB RAM
    1 – 2 GPUs
  - **Special Flavors**



- **Storage**
  Block Storage Volumes
  S3 Object Storage (in progress)

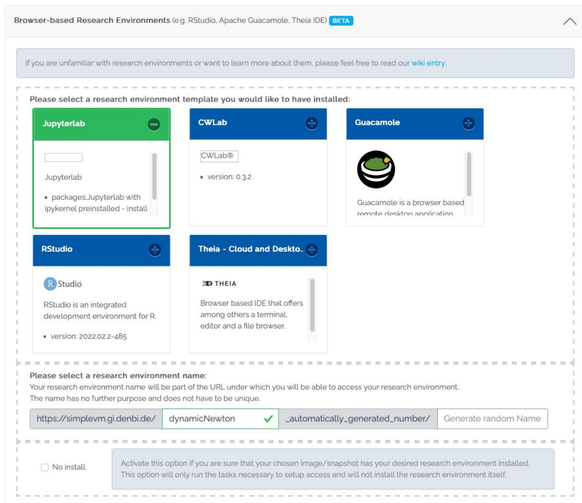
VM Image & Snapshot Management





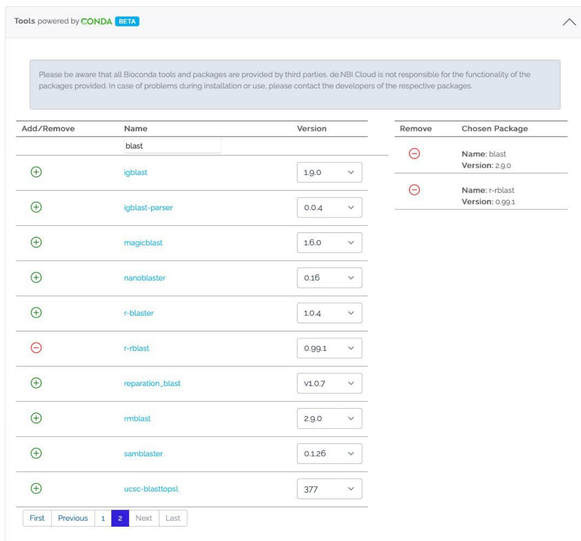
Research Environments

- Pre-configured Applications:
  - RStudio
  - CWLab
  - Guacamole (RDP browser-based desktop access)
  - Theia Cloud IDE
  - JupyterLab



Bioconda Package Management

- Add Bioconda packages to VM image
- Simple dependency management for bioinformatics workflows
- Automatic version and dependency resolution for common bioinformatics tools



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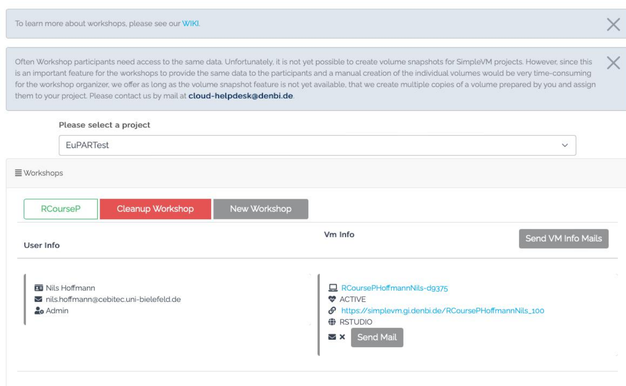
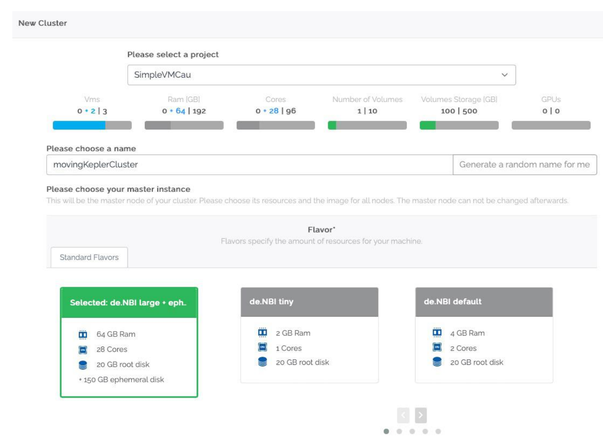
Workshop Support

- Easy setup of workshop participant VMs from image or snapshot
- Assignment of participant to VM, secured via LS AAI
- Access via browser or ssh
- Management of VMs by project administrators



SLURM Grid Cluster

- Based on BiBiGrid (SLURM) cluster
- Automatic bootstrapping
- Master server and worker flavors and images can differ
- Batches support different worker configurations
- Current work on autoscaling



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